

FIELDNODE · CALCULATION

FieldNode sizing calculations

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DOCUMENT CONTROL

Document ID	FND-CAL-001	Version	0.5
Status	Draft	Date	2026-10-02
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/fieldnode		
Source	docs/04-calcs/01-sizing.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-02	Amish Chadha	Requirement table: R11 pinout status after the 2026-10-02 decision (FND-DEC-001); no figure changed
0.4	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.3	2026-09-30	Amish Chadha	Design for construction (FND-DDR-003) applied; bracket, V-block, mass, cost and cell swap with the shield updated; open for Amish's review
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	First issue for TRL 3 (core consumption, energy budget and autonomy, airtime, link budget, store and forward, enclosure temperature and charging window, panel voltage, wind and mounting, installation and service, mass, cost)

FieldNode sizing calculations

With the decisions of FND-DDR-002 applied, FieldNode meets fifteen of its eighteen requirements on paper (ten by calculation, five by design), has two at risk, misses none and leaves one that only a timed installation can settle. Version 0.1 found one miss, heat: in full sun at 45 °C ambient the base enclosure reaches 58.7 °C clean and 66.2 °C dusty on a hot design day, and up to 73.3 °C with the sun in its worst position, against the 60 °C of R3, and the cell is then too hot to charge for most of the day. Amish decided on 2026-09-25 to fit a ventilated white sun shield at hot-climate sites only. This note shows that the shield is needed where the site's design maximum exceeds 30 °C; with it the inside peaks at 48.5 to 52.2 °C at 45 °C ambient and the node stores 10.8 to 13.1 Wh on a hot clear day. The shield is BOM line 14, an option at \$9.00 and 0.16 kg; the hot-climate node costs \$148.00 and weighs 2.61 kg, and R14 now applies to the base node (2.45 kg after the design for construction of FND-DDR-003). The panel is now 9 V class, which keeps 2.42 V of headroom above a typical charger's 5 V minimum input when hot, where the 6 V class panel had none. The published sensor allowance is 100 mW, which gives 5.75 days of autonomy, a 15 % margin; a cold or aged cell still falls short of 5 days (R6 at risk), and the cell will not charge on clear days colder than about -13 °C (R2 at risk). A firmware rule lengthens the reporting interval at SF10 and slower, so airtime stays within 30 s a day on The Things Network. Every number in this note is printed by `docs/04-calcs/sizing.py`; the tag in brackets, for example [A3], is the line of that script's output that carries it.

Safety: These are first-principles estimates for a paper proof of concept. They do not show that the lithium iron phosphate cell, the charger or the pole mounting is safe. Cell temperature, charge lockout and clamp preload must be checked on hardware before any node is left unattended. See FND-PRC-001, Safety.

■ Scope and method

The note checks every requirement in FND-REQ-001 v0.4 against the design in FND-PRC-001 v0.4 and the parametric model `cad/src/model.py`. The script imports the model's `PARAMS` and `derived()` and builds the model once for part volumes, so the enclosure, panel position, bracket, back plate and V-blocks used here are the ones in the STEP files and in drawing FND-DWG-001. It also reads `bom/bom.csv` and `budget_usd` in `project.yaml`. Run it from the repo root with `python docs/04-calcs/sizing.py`; it also writes `docs/04-calcs/results.csv`.

The design case is a node on a 48.3 mm pole with the enclosure base 1.75 m above ground, facing the equator, reporting a 20-byte LoRaWAN uplink every 15 min (DDR-001, D6) to a gateway 2 km away in suburban terrain.

■ Assumptions

Table 1. Main assumptions.

Area	Assumption	Basis
Energy chain	6 W panel; 2 peak sun hours on the tilted panel in the worst month; 20 % panel loss (heat, dust, angle); charger 85 %; cell charge 95 %; rail converters 90 %	As FND-PRC-001 v0.2; screening values
Cell	LiFePO4 3.2 V, 6 Ah; 80 % usable; 70 % of capacity at -20 °C; 80 % at end of life; charging allowed 0 to 45 °C	Typical cell data (see FND-PRC-001)
Core	SX1262 at +14 dBm 45 mA, receive 4.6 mA; two 0.1 s receive windows; controller awake 0.5 s at 8 mA per report; 33 µA whole-node sleep and quiescent current at 3.3 V	Semtech SX1261/2 datasheet figures; the rest assumed
Radio	20-byte payload plus 13 bytes of LoRaWAN overhead; 125 kHz; coding rate 4/5; 8-symbol preamble; explicit header; CRC on	LoRaWAN defaults
Link	+14 dBm; 2 dBi antennas; 0.5 dB node and 2 dB gateway feeder loss; 6 dB receiver noise figure; Okumura-Hata suburban model at 868 MHz; 10 dB fade margin	Screening values; gateway sensitivity assumed equal to the node's
Storage	32 bytes per stored reading; 16 MB SPI flash; 24 bytes per reading when several are batched into one uplink	Assumed record format
Heat	Combined convection and radiation 10 W/m ² K (still air); absorptance 0.45 clean light grey, 0.70 dusty; clear-sky direct beam from the Meinel air-mass model, diffuse 10 % of direct, ground albedo 0.2; back face against the mounting plate; panel shading traced from the model geometry; the ventilated white shield (BOM line 14, 0.5 mm aluminium standing 15 mm off the box) passes a quarter of the solar gain	Handbook ranges; the shield factor matches WWT-CAL-001 and is not measured
Design days	Hot: latitude 28.6° N, late May, clear, 30 to 45 °C. Cold: latitude 50° N, late January, clear, -20 to -10 °C	Chosen to bracket R2
Panel voltage	9 V class panel (FND-DDR-002), Vmp 9.0 V at 25 °C; the 6 V class panel of v0.1 kept for comparison; -0.35 %/K; panel 30 K above ambient in full sun; charger minimum input about 5.0 V	Typical crystalline silicon and bq24650 or LT3652 class figures; to confirm per part at TRL 4
Wind	35 m/s gust; air 1.225 kg/m ³ ; drag coefficient 1.2 on the panel (normal to it) and 1.3 on the enclosure	Screening values, not a code check
Mounting	Band clamp preload 1,000 N; each clamp presses the V-block with twice its preload; friction 0.2; 6063-T5 class flat bar, E = 69 GPa	Preload assumed, to be measured
Mass	Made parts from model volumes (polycarbonate 1.2, aluminium 2.7, ASA 1.07 g/cm ³), shield included from its model volume plus 0.02 kg of fixings; bought parts from typical catalogue masses (panel 0.55 kg, cell 0.15 kg); the 9 V class panel assumed the same mass as the 6 V class	Estimates

■ A. Core consumption and energy budget (R5, R6, R7)

- **Core.** Each report costs 16.0 mA·s at SF9, so 96 reports use 0.43 mAh a day and sleep adds 0.79 mAh: the core needs 4.0 mWh a day [A1]. The TRL 2 figure stands.
- **Worst month.** The panel sees 12.0 Wh, delivers 9.60 Wh, the charger passes 8.16 Wh and the cell stores 7.75 Wh a day [A2].

- **Allowance and autonomy.** The cell holds 15.36 Wh usable. A sensor allowance of 115.0 mW is the largest that gives exactly 5.0 days without sun [A3]; at that load the node draws 3.07 Wh a day and the margin on R6 is zero. At -20 °C the same cell lasts 3.50 days and at end of life 4.00 days [A4].
- **At 100 mW,** the published allowance (FND-DDR-002) and the R7 target, the node draws 2.67 Wh a day, stores 2.9 times that in the worst month, and lasts 5.75 days, a 15 % margin on R6, or 4.03 days at -20 °C and 4.60 days at end of life. After five sunless days it refills in 2.6 worst-month days [A5].
- **Energy neutrality.** The worst month supports sensor loads up to 291 mW before the cell runs down day by day [A6]; autonomy, not harvest, limits the allowance.
- **Sibling loads.** AirStreet (45 mW), NoiseMap (21 mW), FloodGauge (7 mW), WellSense, HeatMap Node and SlopeWatch (about 1 mW or less) all fit within the published 100 mW. CurbCount's 300 mW does not [A7]; this is a cross-repo action in `docs/REVIEW.md`.

■ B. Airtime, link and store and forward (R8, R9, R10)

Table 2. Time on air for a 20-byte uplink (33 bytes on air) at 125 kHz [B1].

Spreading factor	Per uplink	Per day at 15 min	Shortest interval within 30 s/day	EU868 1 % off-time
SF7	71.9 ms	6.9 s	4 min	7 s
SF8	133.6 ms	12.8 s	7 min	13 s
SF9	246.8 ms	23.7 s	12 min	25 s
SF10	452.6 ms	43.5 s	22 min	45 s
SF11	987.1 ms	94.8 s	48 min	99 s
SF12	1,810.4 ms	173.8 s	87 min	181 s

- **R9 is met on paper by a firmware rule.** The 15 min default fits The Things Network's 30 s a day at SF7 to SF9, which covers the R8 design case, but not at SF10 or slower. Under the rule decided in FND-DDR-002, firmware on The Things Network lengthens the interval to the shortest value in Table 2 when adaptive data rate moves a node to SF10 or slower: 22 min at SF10, 48 min at SF11 and 87 min at SF12, which keeps airtime at 30.0 s a day or less [B1b]. On a private TwinKit gateway only the regional duty cycle applies, and the 15 min interval fits it at every spreading factor. The rule is a firmware requirement; the firmware itself is TRL 4 work.
- **Power limit.** At +14 dBm with a 2 dBi antenna the node radiates 15.5 dBm EIRP, 13.35 dBm ERP, inside the 14 dBm ERP limit of EU868; the whip's middle sits 1.64 m above ground [B2].
- **R8 is met on paper.** At SF9 the link budget is 145.0 dB. The Hata suburban path loss at 2 km to a 30 m gateway is 126.4 dB, a margin of 18.6 dB; with a 10 dB fade margin the range is 3.5 km. A 15 m gateway, below Hata's range of validity, gives about 2.5 km. At SF7 the margin is 13.6 dB and the range with fade margin 2.5 km [B3]. Clutter, foliage and gateway height dominate these figures; only a field survey settles them.
- **R10 is met on paper.** A day of readings takes 3,072 bytes; 30 days take 90 kB of the 16 MB flash, which would hold 5,461 days [B4].
- **Forwarding the backlog.** After a 30-day outage the node holds 2,880 readings. At SF9, batching four per uplink, resending them takes 428 s of airtime: 68 days at the 6.3 s a day left under fair use, but half a day under the EU868 duty cycle alone. At SF7 it takes 5 days under fair use [B5]. The store is ample; on The Things Network the backlog is slow to drain, so old readings should be sent when a private gateway is in range or thinned.

■ C. Enclosure temperature and the charging window (R2, R3, R4, R5)

- **Loss coefficient.** The enclosure's exposed faces (all but the back, which sits on the mounting plate) total 0.0930 m² and lose 0.930 W/K. About 852 J/K of enclosure, cell, boards and plate gives a 15 min time constant, so the inside follows the sun within the hour [C1].
- **What the panel hood does.** The panel's front edge overhangs the lid by 56.6 mm and sits 120.7 mm above the enclosure top [C1b]. With the sun 60° high in front it shades 86 % of the top but none of the front face [C2b]. The hood protects the lid seam from rain; it does little for heat when the sun is low or to the side.
- **Worst sun position.** At 45 °C ambient, the worst clear-sky sun position is 35° high and 30° off the front normal: 16.9 W absorbed and 63.3 °C inside when clean, 26.3 W and 73.3 °C when dusty [C2]. This is an upper bound, since the sun is not always in that position at the day's peak ambient. With the shield the same position gives 49.6 °C clean and 52.2 °C dusty [C2c].
- **Where the shield is needed.** Without the shield the dusty box stays at 60 °C or less in the worst sun position up to 31.7 °C ambient, so the shield is fitted where the site's design maximum ambient exceeds 30 °C [C2c]. At that limit, on a clear 15 to 30 °C day, the base node peaks at 44.2 °C clean and 51.2 °C dusty and still stores 19.0 to 25.5 Wh against 2.67 Wh drawn [C3c].

Table 3. Hot design day, 30 to 45 °C, latitude 28.6° N, clear [C3].

Case	Peak inside	Sun hours with charging allowed	Energy stored in the day
Clean, no shield	58.7 °C	2.1 of 13.3 h	0.8 Wh
Dusty, no shield	66.2 °C	1.4 of 13.3 h	0.3 Wh
Clean, with shield (hot-climate option)	48.5 °C	7.4 of 13.3 h	13.1 Wh
Dusty, with shield (hot-climate option)	50.4 °C	6.5 of 13.3 h	10.8 Wh

- **R3 is met on paper with the shield at hot-climate sites.** Without the shield the clean enclosure scrapes under 60 °C on the 45 °C design day but not with dust or with the sun in its worst position, which is why R3 now requires the shield where the design maximum exceeds 30 °C (FND-REQ-001 v0.4). The result rests on the assumed shield factor of 0.25, which only a test can confirm.
- **Hot weather stops charging.** Because the 45 °C charge lockout (R4) acts on a cell that sits above 45 °C from mid-morning, a clean node stores 0.8 Wh on a hot clear day against 2.67 Wh drawn. A run of such days loses 1.84 Wh a day, so a full cell lasts 8.3 days (6.4 days dusty). With the shield the day ends 10.4 Wh in credit (8.1 Wh dusty) [C3b]. The lockout is correct and must stay; the fix, now decided, is to keep the cell cooler with the shield at hot sites.
- **Cold weather.** On a clear day at 50° N with -20 to -10 °C ambient, sun on the box lifts the inside to 3.3 °C, which allows 4.2 h of charging and 8.7 Wh stored [C4]. The peak rise is 13.3 K, so on clear days whose maximum is below about -13 °C the cell never reaches 0 °C and cannot charge [C5]. Overcast cold spells longer than the cold autonomy of about 4 days (section A) will stop the node.
- **Panel voltage.** In full sun on the hot day the panel runs at about 75 °C. The 9 V class panel decided in FND-DDR-002 then gives a Vmp of 7.42 V, 2.42 V above a typical 5.0 V charger minimum input; the 6 V class panel of v0.1 fell to 4.95 V, 0.05 V short [C6]. The part choice for the power board (TRL 4 work, on hold) must confirm the minimum input and the charger's maximum input.

- **R2 is at risk at the cold end.** With the shield at hot sites the warm end is covered (52.2 °C dusty worst case at 45 °C, and 58.3 °C for an unshielded node at 30 °C, both under the 70 °C electronics rating), but the cell charges only on clear days warmer than about -13 °C.

■ D. Wind and mounting (R12, R13)

- **Loads.** A 35 m/s gust gives a dynamic pressure of 750 Pa: 52.2 N normal to the panel, 29.3 N on the enclosure front-on and 17.6 N side-on [D1].
- **Overturning.** With wind from behind, the panel lifts and pushes forward; together with the node's weight this gives 14.8 N·m about the lower clamp and pulls the upper clamp off the pole with 59 N. Wind from the front presses the node onto the pole [D2]. Against the 2,000 N that a 1,000 N band preload holds, the factor is 33.8 [D3]. With the shield fitted the enclosure presents 36.3 N front-on and the worst clamp pull rises to 63 N, a factor of 31.7 [D3b].
- **Slip.** Weight and wind push the node down the pole with 64 N against 800 N of friction (factor 12); side wind twists it with 1.58 N·m against 19.3 N·m (factor 12) [D4]. These factors depend on the assumed preload, which a torque check at installation must confirm.
- **Bracket.** Since FND-DDR-003 each side has a post, 161 mm between its lower foot bolt and its head bolt at 80°, and a strut, 129 mm between bolts at 38°, bolted flat to angle clips on the back plate and on the panel frame's back lip. Each member carries at most about 39 N, and a 20 x 3 mm flat bar of that length buckles out of plane at 1,180 N, a factor of 30 [D5]. The post is held by two bolts at its foot, so the side frame is a rigid triangle, not a linkage.
- **Pole range.** Each V-block is 60 x 33 x 20 mm with a true 90° V 50.3 mm wide at its face; the 48.3 mm design pole touches both V faces 24.9 mm from the plate [D6b]. On a 40 mm pole the V contacts sit 14.1 mm either side of the center and on a 60 mm pole 21.2 mm, both inside the V, which seats poles up to 71 mm. Each band needs about 229 to 280 mm of length around the pole, block and plate [D6], [D6b].
- **R13 is met on paper,** subject to the preload assumption. The pole, its footing and any wall anchors are site supplied and outside this note; the node adds about 81 N of wind load about 2 m up the pole, which the site owner should check.

■ E. Installation and service (R12, R15)

- **Installation.** Fitting the bracket and panel on the ground, lifting the node, tightening two clamps, aiming the panel, plugging in the sensor and confirming an uplink take an estimated 15 min with a helper [E1], exactly the R12 limit. Only a timed installation can verify it.
- **Cell swap.** Opening the lid, swapping the cell in its fused holder, renewing the desiccant and confirming an uplink take an estimated 7 min with a screwdriver and no soldering [E2]. Where the sun shield is fitted it covers the lid; it lifts off after four thumb screws, which adds about 2 min, 9 min in all [E2b]. R15 is met by design for both.

■ F. Mass and cost (R14, R16)

- **Mass.** The enclosure body, lid and lugs weigh 0.43 kg, the printed ASA internal plate 0.07 kg, the bracket 0.22 kg and the back plate with V-blocks 0.48 kg; bought parts add 1.24 kg; 2.45 kg in all for the base node [F1]. The TRL 2 figure of about 1.7 kg left out most of the back plate. The design for construction (FND-DDR-003) added clips, bolts, a connector strip and fuses, and took mass out with a window in the back plate behind the enclosure and a 20 mm bar. R14, stated for the base node, is met on paper with 0.05 kg of margin; the margin is thin and rests on

catalogue masses. The shield, 181 x 106 x 206 mm of 0.5 mm aluminium with its fixing flanges, weighs 0.16 kg with its thumb screws, so a hot-climate node weighs 2.61 kg [F1b].

- **Cost.** The BOM has 15 lines, all priced; line 14, the shield, is an option at quantity 0 in the base node, and line 15, the plug-in connectors and rail fuses, was added by FND-DDR-003. The base node totals \$139.00 against the \$150 value-engineering target (`budget_usd` , a hypothetical control target), \$11.00 (7 %) under [F2]. A hot-climate node with the shield costs \$148.00 and weighs 2.61 kg [F3]. R16 is met on paper for both, within the value-engineering target.

■ **L. Results against every requirement**

Table 4. Requirement status from this note [L].

ID	Requirement	Value	Target	Status
R2	Operating temperature	52.2 °C dusty worst case with the shield at 45 °C; 58.3 °C unshielded at 30 °C; no charging on clear days below about -13 °C	-20 to +45 °C ambient; electronics -20 to +70 °C inside	At risk (cold charging)
R6	Autonomy without sun	5.75 days at the published 100 mW (15 % margin); 4.03 days at -20 °C; 4.60 days at end of life	5 days or more	At risk (cold or aged cell)
R12	Mounting and install	Seats 40 to 71 mm poles; installation estimated at 15 min	40 to 60 mm poles and walls; 15 min or less	Not verifiable at TRL 3 (time); fit met by design
R3	Interior temperature in sun	With the shield at 45 °C: 48.5 °C clean, 50.4 °C dusty on the hot day, 52.2 °C worst sun position; unshielded at 30 °C: 58.3 °C worst sun position (at 45 °C: 58.7 to 73.3 °C)	60 °C or less; shield where the design maximum exceeds 30 °C	Met on paper (shield factor assumed)
R5	Energy neutral in the worst month	7.75 Wh stored against 2.67 Wh drawn; 10.8 to 13.1 Wh stored on a hot clear day with the shield; 9 V class Vmp headroom 2.42 V when hot	Harvest exceeds demand at 2 peak sun hours	Met on paper
R7	Sensor power allowance	Published allowance 100 mW; 115.0 mW would give exactly 5 days	100 mW or more	Met on paper
R8	Radio link	18.6 dB margin at 2 km, SF9, 30 m gateway; 3.5 km with 10 dB fade margin	2 km suburban at SF9 or faster	Met on paper
R9	Airtime within fair use	23.7 s a day at SF9 and 15 min; firmware rule gives 22 min at SF10 and 87 min at SF12; 30.0 s a day or less	30 s a day or less on The Things Network	Met on paper (firmware rule)
R10	Store and forward	30 days in 90 kB of 16 MB	30 days or more	Met on paper
R11	Standard sensor interface	Two M12 5-pin ports: switched rail, ground and three signal pins	Two sealed ports with a bus, analog input and one switched rail each	Met on paper (pinout decided as the proposed standard, awaiting sign-off; FND-DEC-001)
R13	Wind	Clamp pull 59 N (63 N with the shield) against 2,000 N; slip factor 12; bracket factor 30	35 m/s without loosening	Met on paper (preload assumed)
R14	Mass	Base node 2.45 kg (0.05 kg margin); 2.61 kg with the shield, which R14 excludes	2.5 kg or less, base node	Met on paper
R16	Cost	\$139.00 base node; \$148.00 with the shield	\$150 value-engineering target for the FieldNode core	Met on paper, within the target
R1	Weather protection	IP65 box and ePTFE vent; IP67 class glands, capped ports and bulkhead	IP65; sealed penetrations	Met by design
R4		NTC gates the charger at 0 and 45 °C		Met by design

ID	Requirement	Value	Target	Status
	Safe charging window		Charging blocked outside 0 to 45 °C	
R15	Serviceable	Cell swap in about 7 min; 9 min with the shield	10 min or less, no soldering	Met by design
R17	Open and independent	CERN-OHL-S-2.0 and MIT; standard LoRaWAN	Open files, any network server	Met by design
R18	Privacy pass-through	Core forwards only what the sensor firmware passes	No images or audio leave the node	Met by design

Counts: none not met, 2 at risk, 10 met on paper, 5 met by design, 1 not verifiable at TRL 3. In v0.1 the counts were 1 not met (R3), 5 at risk (R2, R5, R6, R9, R14) and 6 met on paper; R3, R5, R9 and R14 moved to met on paper because of the decisions in FND-DDR-002. R3 and R5 now rest on an assumed shield factor that only a test can settle.

■ Checks against the TRL 2 figures

TRL 2 claim (FND-PRC-001 v0.2)	This note	Action
7.8 Wh/day stored; 3.1 Wh/day drawn	7.75 and 3.07 Wh/day at 115 mW	Stands
Sensor allowance about 115 mW	115.0 mW gives exactly 5.0 days; 100 mW published (FND-DDR-002)	Precis updated
Autonomy about 5.0 days	5.00 days at 115 mW, 5.75 days at 100 mW; 3.5 to 4.6 days cold or aged	Precis updated
Core about 4 mWh/day	4.0 mWh/day	Stands
Airtime 24 s at SF9, 43 s at SF10	23.7 s and 43.5 s	Stands
Link budget about 145 dB	145.0 dB; 18.6 dB margin at 2 km	Precis updated
Wind about 52 N, about 16 N·m	52.2 N; 14.8 N·m about the lower clamp	Precis updated
Mass about 1.7 kg	2.45 kg	Precis updated; back plate was under-counted
Cost about \$126	\$139.00	Precis updated; parts added for construction (FND-DDR-003)
Panel hood "helps" with interior heat	Shades 86 % of the top but not the front; R3 not met without a shield; charging blocked in hot weather	Precis updated; shield option for sites above 30 °C (FND-DDR-002)
One gland fitted, one blanked	The panel lead needs a gland; both glands are fitted	BOM line 3 updated